

Elaine Hong

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

B.S. in Computer Science and B.S. in Operations Research and Information Engineering

Expected May 2028

Relevant Coursework: Computer Architecture, Computer System Organization, Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics, Linear Optimization, Differential Equations

Activities: NCAA Division I Varsity Fencing, Society of Women Engineers at Cornell E-Board, C.ORIE E-Board

EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

FinMD

Miami, Florida

- Architected a live multi-product analytics dashboard spanning **SCOPE, PRISM, and RADAR**, migrating from mock data to tenant-scoped production metrics with server-side aggregation, improving data reliability
- Designed and shipped a physician identity pipeline **across four product surfaces**, authoring PostgreSQL functions for deterministic identity resolution, stage derivation, and full status history
- Strengthened API security by enforcing **row-level security policies and ownership validation**, offloading filtering to server-side pagination and reducing p50 page load latency by **60%**

Machine Learning Engineer Intern

June 2025 – August 2025

Anote AI

New York, NY

- Conducted a **7-condition RAG ablation** across 133 FinanceBench questions, finding that no standard retrieval strategy outperformed the baseline, blocking underperforming pipelines from production
- Exposed a **10–16×** underestimation of numeric accuracy in ROUGE-L by implementing semantic chunking, achieving **34.8%** numerical Exact Match and motivating improved evaluation of numerical responses
- Built React outreach automation with OAuth and adaptive rate-limiting for high-throughput campaign execution

Undergraduate Research Assistant

August 2025 – June 2026

Cornell Engineering Undergraduate Research

Ithaca, NY

- Built scalable LLM evaluation infrastructure benchmarking **5+ frontier models** against Stockfish across **10,000+ chess positions**, enabling large-scale analysis of reasoning quality and failure modes
- Designed ablation studies across **20+** prompting strategies over **200+** games, identifying recurring failure modes

PROJECTS

AI Asset Generation Platform | *Python, Next.js, Stable Diffusion, REST APIs*

- Built a full-stack AI asset generation platform using Stable Diffusion and Python, automating art production workflows and increasing throughput by **40%**
- Implemented backend generation pipelines with task batching, caching, and asynchronous request handling, reducing average image render time by **30%**

Bear Escape (BigRed//Hacks Winner) | *React, TypeScript, Python, Gemini API, OpenCV, SpaceTimeDB*

- Won **Best Use of Gemini and Best UI** out of **200+** participants for an AI-powered multiplayer rhythm game
- Built synchronized gameplay with Gemini-generated beatmaps, OpenCV motion controls, and SpaceTimeDB

LEADERSHIP

Conference Director | *Society of Women Engineers at Cornell*

Jan 2026 – Present

- Lead conference programming and logistics, expanding professional opportunities for women in engineering
- Coordinate attendees, outreach, and travel planning for 20+ students attending the annual SWE conference

Publicity Chair | *Cornell Operations Research and Information Engineering Club (C.ORIE)*

Jan 2026 – Present

- Lead publicity and outreach for academic, professional, and community programs serving Cornell ORIE students

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, Java, SQL, C

Frameworks & AI: React, Next.js, FastAPI, LangChain, RAG Pipelines, LLM Evaluation, Pandas, NumPy, OpenCV

Data & Infrastructure: PostgreSQL, MySQL, Supabase, Docker, AWS EC2, REST APIs, WebSockets, OAuth, Linux